Rail Alignment Automation with Machine Learning

3D Scanning | Automated Classification | CMAA Compliance Analysis

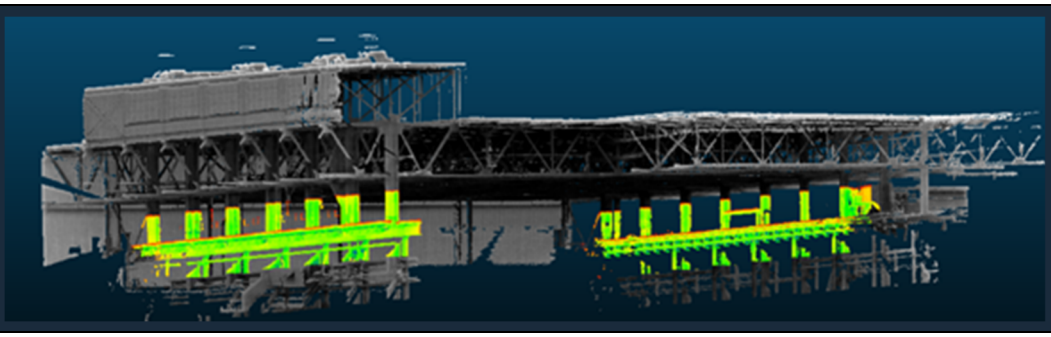
3D Scanner Approach to Crane Rail Survey

Coffman Engineers has developed an advanced rail alignment survey methodology that pairs **3D laser scanning** with **machine learning classification** to deliver faster, more accurate crane rail straightness, elevation, and span analysis.

Using a 3D scanner, we capture millions of XYZ data points eliminating the low-resolution, error-prone process of traditional surveys that only collect a single point at large increments.

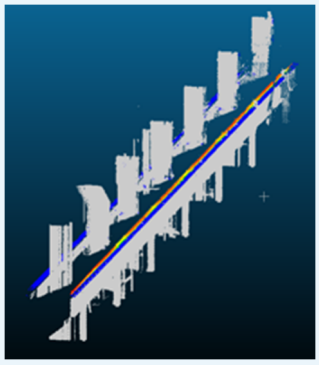
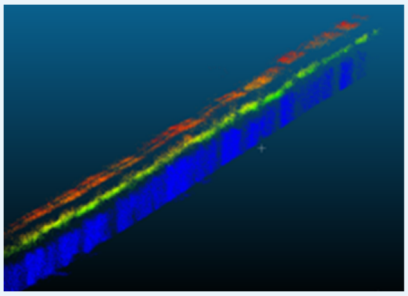
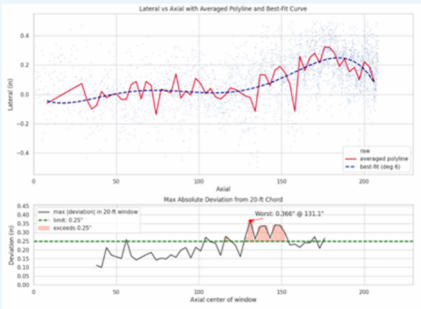
Our machine learning pipeline then classifies rail components from the scan data and produces CMAA-standard alignment analysis with no manual point selection required.

3D Scanner Example: Leica BLK360



3D point cloud capture of crane rail system with rail area of interest highlighted

How It Works

1. Scan & Capture	2. Classify & Isolate	3. Analyze & Report
		
High-density 3D laser scans capture millions of points at small increments from scanner positions on the crane, structure, or ground.	Machine learning models (Random Forest + clustering) isolate rail head, web, and base from the point cloud.	Straightness, elevation, and span analysis against CMAA standards with deviation reporting and adjustment recommendations.

Improved Accuracy	Automation	CMAA Compliance
Small point spacing vs. traditional methods takes advantage of statistical filtering and eliminate human error.	No manual point selection and ML classify rail components and generates adjustment data automatically.	Deviation analysis measured directly against CMAA crane rail standards.